Exercise 1 Q1 Solution

1(a)

Year	Y_t	Naive	3-point M.A.	
2005	1365.8			
2006	1435.9	1365.8	1271.8	($\hat{Y}_{06}$)
2007	1521.4	1435.9	1361.1	($\hat{Y}_{07}$)
2008	1629.2	1521.4	1440.1	($\hat{Y}_{08}$)
2009	1733.7	1629.2	1528.8	($\hat{Y}_{09}$)
2010	1807.2	1733.7	1628.1	($\hat{Y}_{10}$)

$$\hat{Y}_{06} = \frac{1167.9 + 1281.7 + 1365.8}{3} = \frac{3815.4}{3} = 1271.8$$

$$\hat{Y}_{07} = \frac{1281.7 + 1365.8 + 1435.9}{3} = \frac{4083.4}{3} = 1361.1$$

$$\hat{Y}_{08} = \frac{1365.8 + 1435.9 + 1521.4}{3} = \frac{4323.1}{3} = 1441.0$$

$$\hat{Y}_{09} = \frac{1435.9 + 1521.4 + 1629.2}{3} = \frac{4586.5}{3} = 1528.8$$

$$\hat{Y}_{10} = \frac{1521.4 + 1629.2 + 1733.7}{3} = \frac{4884.3}{3} = 1628.1$$

Data for trend and causal models

Year	t	Y_t (retail sales)	X_t (disposable income)
1990	1	375.2	710.0
⋮	⋮	⋮	⋮
2005	16	1365.8	2832.1

Summary statistics

$\sum t = 136$

$\sum t^2 = 1496$

$\sum t \cdot Y_t = 131771.5$

$\sum Y_t = 12850.6$

$\sum Y_t^2 = 11839206.12$

$\sum X_t = 25538$

$\sum X_t^2 = 48108536.48$

$\sum X_t Y_t = 2384559.54$

The above summations are on t going from 1 to 16. That is

$$\sum_{t=1}^{16}$$

1. (a) (cont'd)

Trend model (based on time t)

Slope estimate:

$$\hat{\beta} = \frac{\sum tY_t - \frac{(\sum t)(\sum Y_t)}{16}}{\sum t^2 - \frac{(\sum t)^2}{16}} = \frac{131771.5 - \frac{136(12850.6)}{16}}{1496 - \frac{(136)^2}{16}}$$

$$= \frac{2254.14}{340} = 66.2982 \simeq 66.3$$

Y-intercept estimate:

$$\hat{\alpha} = \bar{Y}_t - \hat{\beta}\,\bar{t} = \frac{12850.6}{16} - 66.2982\left(\frac{136}{16}\right)$$

$$= 803.1625 - 563.5347 = 239.6278 \simeq 239.63$$

$\therefore$ The trend model is

$$\hat{Y}_t = 239.63 + 66.3t, \quad t = 1, 2, \ldots, 21.$$

Causal model (based on X_t, the regressor or independent variable)

$$\hat{\beta} = \frac{\sum X_t Y_t - \frac{(\sum X_t)(\sum Y_t)}{16}}{\sum X_t^2 - \frac{(\sum X_t)^2}{16}} = \frac{23845579.64 - \frac{25538(12850.6)}{16}}{48108536.48 - \frac{(25538)^2}{16}}$$

$$= 0.4539 \simeq 0.454$$

$$\hat{\alpha} = 803.1625 - 0.453866\left(\frac{25538}{16}\right) = 78.7356 \simeq 78.7$$

$\therefore$ The causal model is

$$\hat{Y}_t = 78.7 + 0.454X_t, \quad t = 1, 2, \ldots, 21.$$

1(b)

$$e_t = Y_t - \hat{Y}_t$$

$$MAE = \frac{\sum_{t=2006}^{2010} |e_t|}{5} \qquad MAPE = \frac{\sum_{t=2006}^{2010} \frac{|e_t|}{Y_t}}{5}$$

$$MSE = \frac{\sum_{t=2006}^{2010} e_t^2}{5} \qquad RMSE = \sqrt{MSE}$$

N.B. Forget about the unbiased estimate of MSE for the time being.

Year	t	Y_t	$\hat{Y}_t$ (trend)	$\lvert e_t \rvert$	e_t^2	MAPE
2006	17	1435.9	1366.73	69.17	4784.4889	0.04817
2007	18	1521.4	1433.03	88.37	7809.2569	0.05808
2008	19	1629.2	1499.33	129.87	16866.2169	0.07971
2009	20	1733.7	1565.63	168.07	28247.5249	0.09694
2010	21	1807.2	1631.93	175.27	30719.5729	0.09698
				630.75	88427.0605	0.37988

X_t	$\hat{Y}_t$ (Causal)	$\lvert e_t \rvert$	e_t^2	MAPE
3007.6	1444.1504	+8.2504	68.0691	0.00575
3184.2	1524.3268	+2.9268	8.5602	0.00192
3467.9	1653.1266	+23.9266	572.4822	0.01469
3710.0	1763.04	+29.34	860.8356	0.01692
3949.1	1871.5914	+64.3914	4146.2524	0.03563
		128.8352	5656.2055	0.07491

1(c)

	Trend	Causal
MAE	126.15	25.7670
MAPE	0.07597	0.01498
MSE	17685.4121	1131.2411
RMSE	132.9865	33.6339

Causal model is much better! It can be seen from the calculations of the error terms, causal model's errors are much smaller than the trend model.